

Three metres of cushion, for free

A dam that opened its gates three days late, replayed against the schedule it could have used instead.

CLIMATE RESILIENCE & DISASTER PREPAREDNESS

IDUKKI + IDAMALAYAR

18 RESERVOIRS INGESTED

ADVISORY ONLY

REPLAY ACCURACY 0.30_m Mean level error over 20 days of October 2021, against the KSEB bulletin.	CUSHION GAINED ~3_m Extra freeboard under the optimised release schedule.
REVENUE EFFECT +4–10_{cr} Positive. Same water, shifted into higher-tariff hours.	OPTIMISER'S TARGET 728.43_m Against KSEB's published rule level of 728.50 m. A 7 cm gap.

01 – THE GAP

The decision is made from the past, not the forecast

A release decision at a Kerala dam is made from the current reservoir level and the rain that has already fallen. It is not made from a simulation of where the reservoir will be in seventy-two hours.

That single fact sets everything else. If you only act once the level is high, you act late – and because you act late, you must act big. The gates that open on day four open wide, because the cheap option is no longer on the table.

Two gaps sit underneath it. There is **no forward-looking tool** that turns a rainfall forecast into a defensible release schedule, so an operator has no artefact to point at when justifying a release while the sky is still clear – and releasing water that is not yet needed looks, to an auditor, like wasting it. And there is **no joint scheduling across a cascade**, so two individually reasonable decisions on the same river arrive downstream as one surge.

Catchment response makes this sharper than it sounds. Kirpich lag for the steep Western Ghats catchments feeding Idukki is well under a day. By the time inflow is visibly climbing, the decision window has already shut.

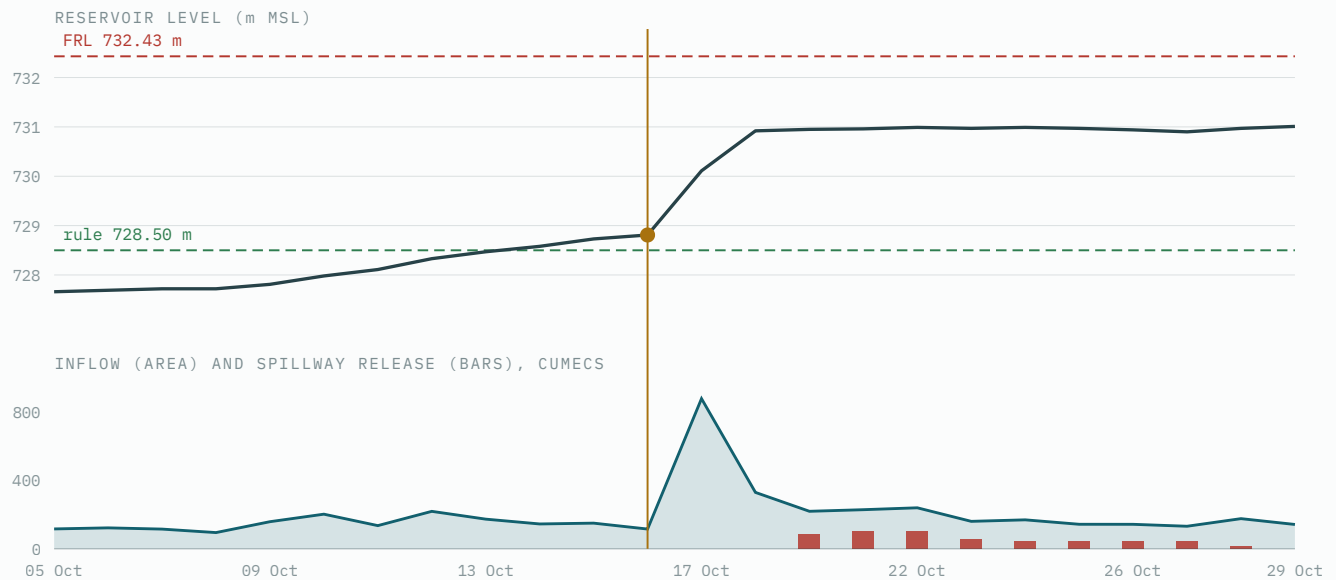
The people downstream get **hours** of warning where the physics allowed **days**.

02 – THE EVENT

October 2021, day by day

Every value below is the published KSEB bulletin reading for Idukki. Drag the scrubber and watch the decision window close.

Idukki reservoir · 5–29 October 2021



● 0.31 m above rule level, spillway shut. Cushion remaining to FRL: 3.62 m.

The shape of the failure is visible without any model at all. On **16 October** the reservoir stood at 728.81 m – already above its own published rule level of 728.50 m – with the spillway shut. The next day brought 168 mm of rain and inflow rose **7.6× in twenty-four hours**, from 115.7 to 879.2 cumecs.

The gates opened on **20 October**, at 730.95 m. Three days after the surge, and two metres higher than the point at which acting would have been cheap.

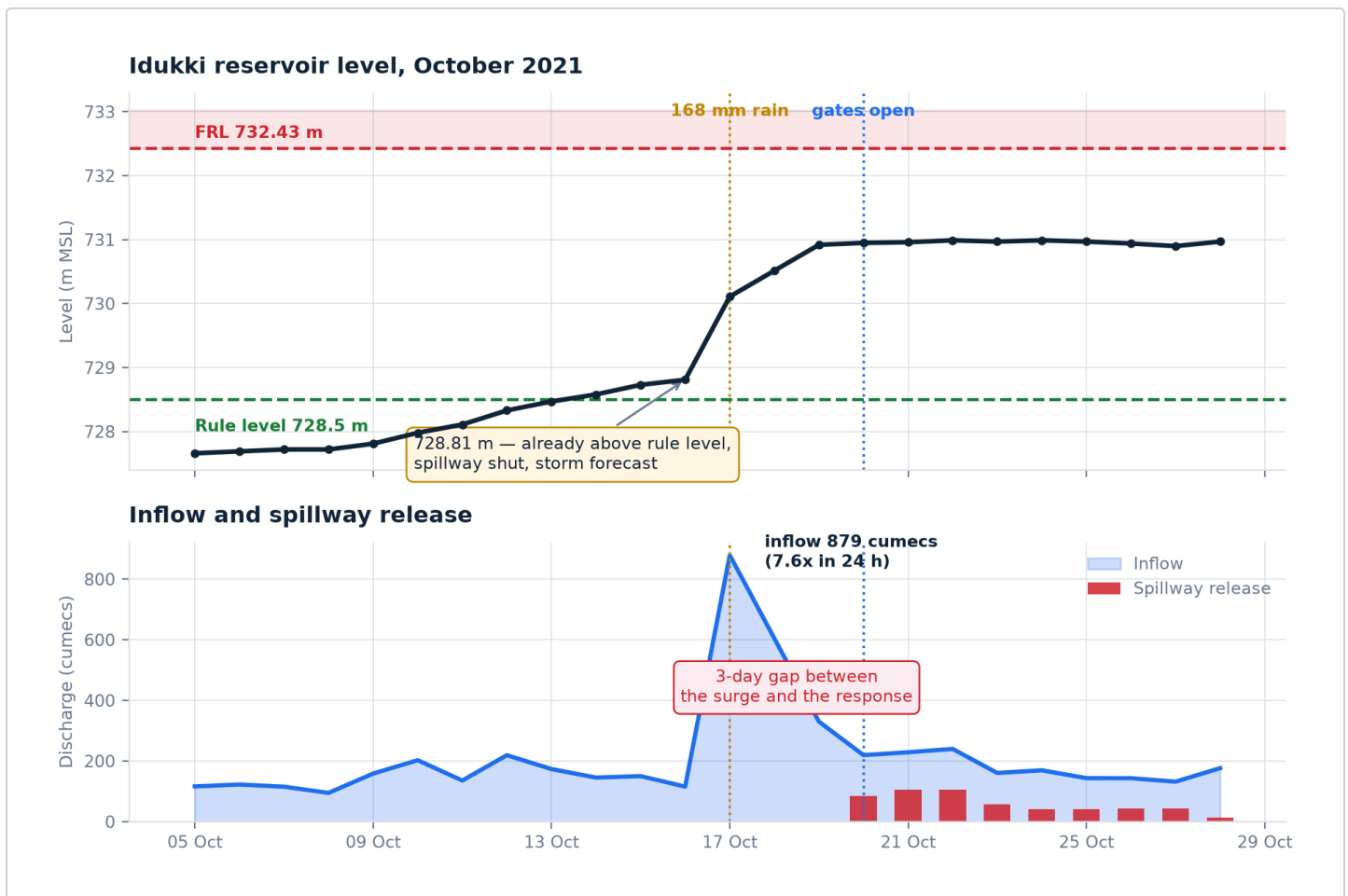


Figure 1 – the three-day gap. Level crosses the rule line on 12 October and keeps climbing. The spillway stays shut through the entire 168 mm rain day and the inflow peak that follows it. Response begins on the 20th, once the reservoir has already absorbed the surge.

03 – THE CASCADE

Two dams, one river, the same day

Idukki is not alone on the Periyar. Nothing coordinates it with Idamalayar, and on 20 October 2021 both opened their spillways.

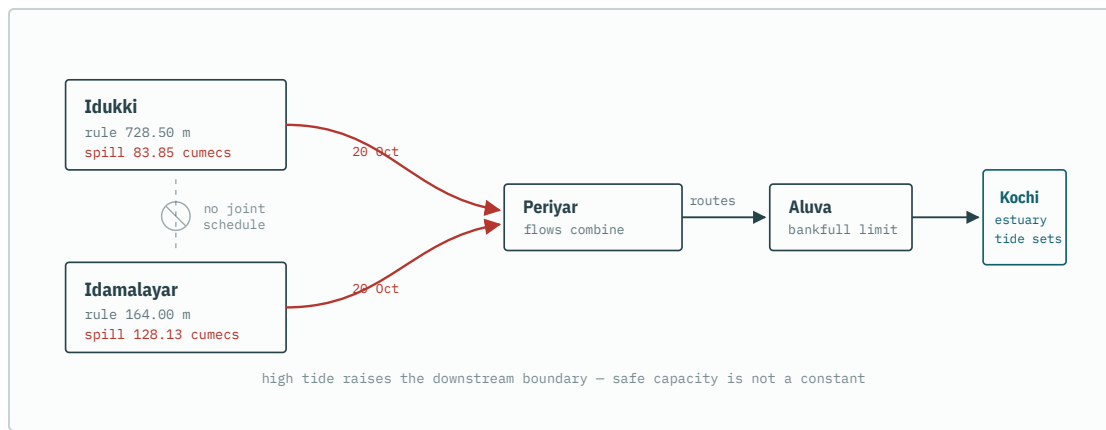


Figure 2 – the missing edge. Each dam's decision is locally defensible. What no one owns is the sum of them at Aluva, arriving on a river whose safe capacity moves with the tide at Kochi. The crossed link is the gap AquaSync's cascade layer closes.

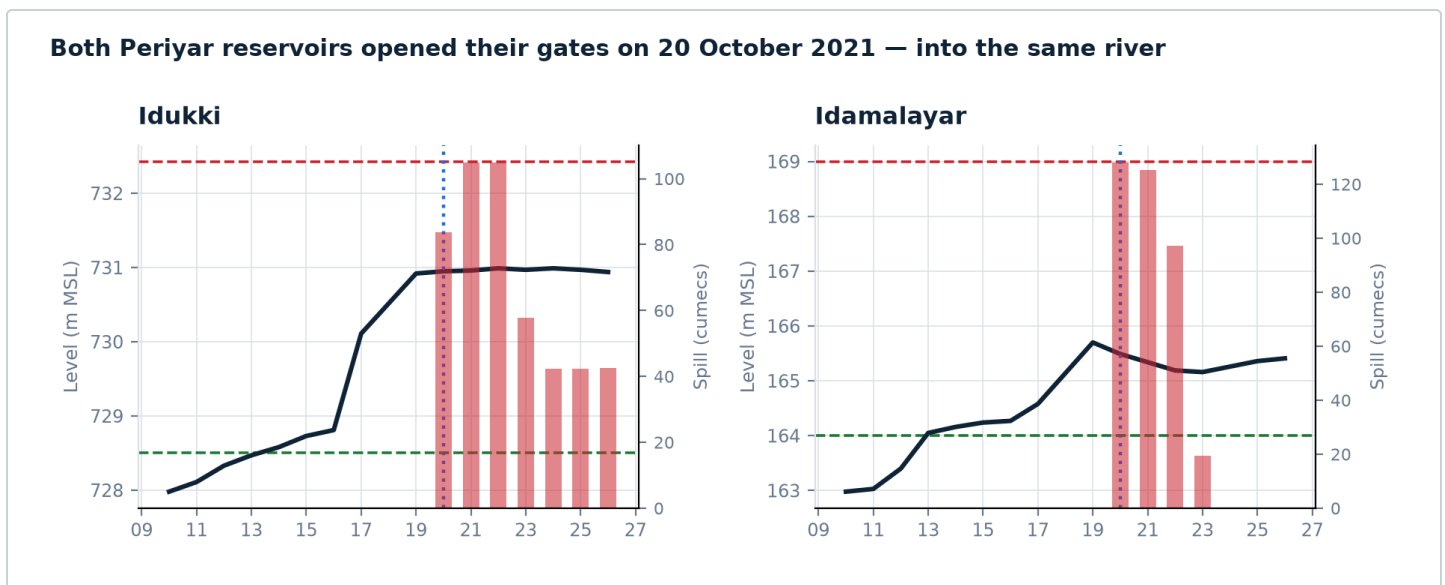


Figure 3 – both dams, same window. Two independent operators, both acting reasonably by their own rule curve, produce a combined release the downstream reach was never scheduled to absorb.

04 – METHOD

Four layers, five models, one recommendation

Ingestion feeds simulation; simulation feeds a policy search; the search hands a schedule to a human who approves or rejects it. The simulation core is pure NumPy and imports nothing from the API – that is what makes it testable.

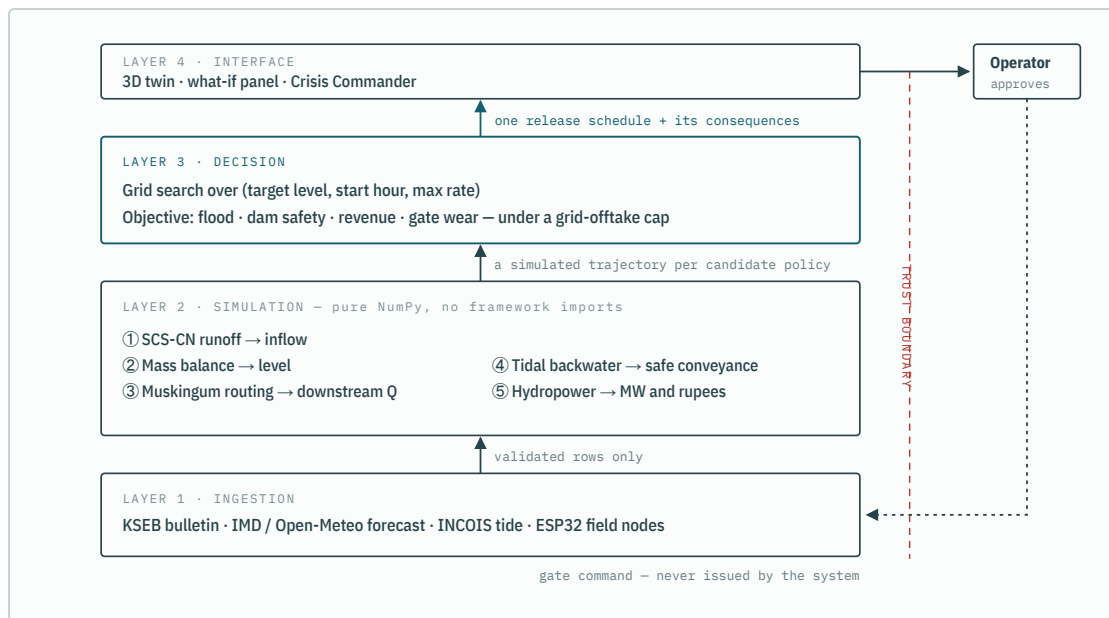


Figure 4 — where the recommendation stops. Every arrow crossing the dashed boundary is a human decision. The system computes and presents; it does not act. That line is permanent, not a phase-one limitation.

① RUNOFF

Rainfall becomes inflow

$$Q = (P - 0.2S)^2 / (P + 0.8S)$$

SCS Curve Number for losses, triangular unit hydrograph for timing. The curve number shifts across AMC-I/II/III on a rolling five-day antecedent depth that updates as the storm proceeds – a long spell saturates its own catchment and the runoff coefficient climbs hour by hour. That feedback is what turns a wet week into a flood.

② ROUTING

Release becomes downstream flow

$$\text{stable iff } 2Kx \leq dt \leq 2K(1-x)$$

Muskingum storage routing, Muskingum–Cunge where K and x must come from channel geometry. A 38 km reach with $K = 8$ h at an hourly step violates the lower bound and the hydrograph oscillates below zero – the river appears to run backwards. Reaches are auto-split into N sub-reaches of K/N .

③ POWER

Water becomes rupees

$$P = \rho g Q H \eta(Q, H) \cdot \text{ToD tariff bands}$$

Hill-diagram efficiency rather than a flat coefficient, against KSEB time-of-day tariff bands. Verified at 743 MW for rated flow against a 780 MW nameplate. This is the model that turns "release early" from a pure cost into a schedulable asset.

④ RESERVOIR

Inflow becomes level

$$dV/dt = Q_{in} + Q_{rain} - Q_{turb} - Q_{spill} - Q_{evap}$$

Hourly explicit Euler. Level $\leftrightarrow$ storage via a fitted power law $S(h) = S_{frl} \cdot ((h - h_{dead}) / (h_{frl} - h_{dead}))^\beta$, because no surveyed elevation–capacity table exists. $\beta = 1.348 > 1$, which is precisely why the top two metres are worth more as cushion than the bottom twenty.

⑤ TIDE

The sea sets the boundary

$$\eta(t) = \sum A_i \cos(\omega_i t - \phi_i) \cdot M2 \ S2 \ K1 \ O1 \ N2$$

Harmonic prediction for Kochi plus exponential backwater decay upstream from the mouth. A release that is safe at low tide is not safe at high tide, so the downstream constraint is a moving target the schedule has to respect. Runs entirely offline – no API, no network.

◆ SEARCH

The decision layer

$$\text{argmin over (target, start hour, max rate)}$$

Exhaustive deterministic grid – no stochastic optimiser, so the same inputs always give the same schedule and the result is auditable. Constrained by what the grid will actually take: without that cap the optimiser books revenue no one would ever buy.

A model that has not been tested is a hypothesis

Two things anchor every result in this document: a calibrated level–storage curve, and a replay of October 2021 that reproduces what actually happened.

Catching the 11% that would have broken the model

The public dam dataset reports, between 25 September 2020 and 30 April 2021, live storage above the reservoir's physical capacity and storage percentages over 1,000%. About **11% of rows are physically impossible**.

This is not a cosmetic data-cleaning note. Fitting the level–storage curve on the raw feed gives $r^2 = 0.784$; on validated rows, $r^2 = 0.9957$. The corrupt block alone displaces the curve by **more than the flood cushion being modelled** – a twin fitted on the raw feed would be confidently wrong in exactly the region where the decision is made.

LEVEL-STORAGE FIT	RAW FEED	VALIDATED ROWS
Rows used	2,056	1,836
β – curve exponent	1.302	1.348
r^2	0.784	0.9957
Mean absolute error	174 Mm ³	17 Mm ³
MAE as share of mean storage	19%	1.9%

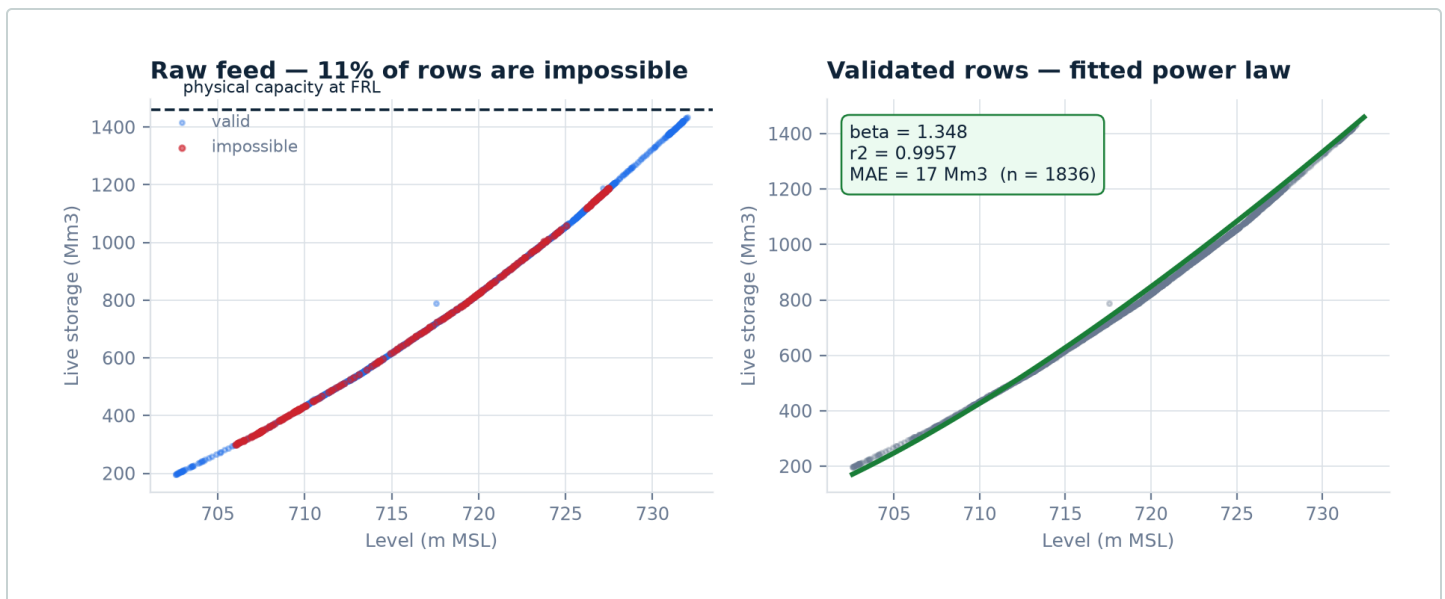


Figure 5 – the validation layer, justified. The raw feed carries the corrupt 2020-09 → 2021-04 block, which drags the fit; the same curve on validated rows lands at $r^2 = 0.9957$. The gap between the two is the entire argument for building an ingestion validator instead of trusting the source.

The replay

Fed the releases that actually occurred over 481 hourly steps of October 2021 – with no fitting to the level series itself – the twin tracks observed reservoir level to 0.30 m mean error, 0.57 m maximum. Against a reservoir that moved more than three metres over the window, the twin reproduces the episode closely enough to plan a decision against.

06 – LEAD TIME

What earlier warning actually buys

The intuition is that more forecast lead time means more flood cushion. The study says otherwise – and the real answer is more interesting.

Lead-time explorer · energy-neutral policies

CUSHION GAINED

3.28_m

RELEASE SPILLED

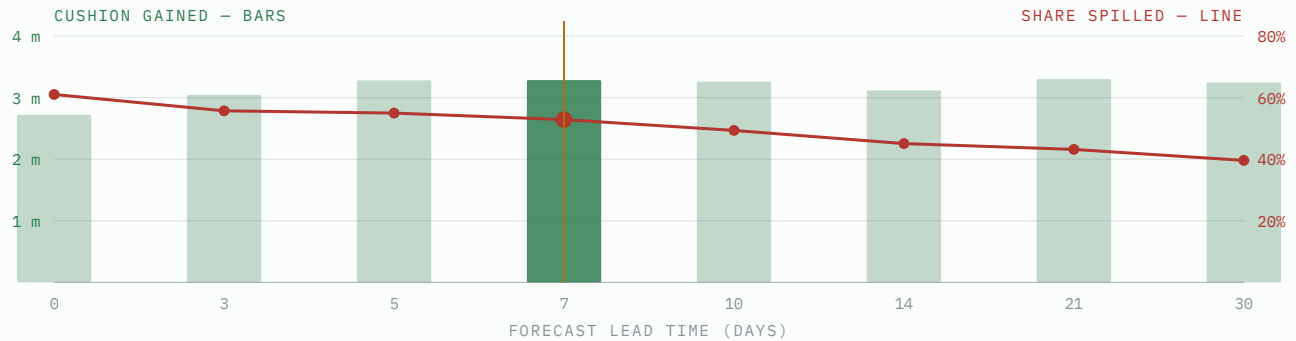
53%

REVENUE DELTA

+9.6_{cr}

START LEVEL

727.81_m



SCHEDULE THE OPTIMISER RETURNS

From hour 118, release up to 828 cumecs until Idukki reaches 728.43 m, then hold within ± 0.15 m.

Cushion is **flat**. Roughly three metres at zero days of warning and roughly three metres at thirty. Acting on the day is nearly as good as acting a month out, for safety.

What lead time changes is **waste**. At zero lead, 61% of released water goes over the spillway and earns nothing. At thirty days, 40% – the rest goes through the turbines. The safety-versus-power trade-off everyone assumes is largely an artefact of **hoarding reservoir level instead of scheduling releases**. Once you schedule, the trade-off mostly dissolves.

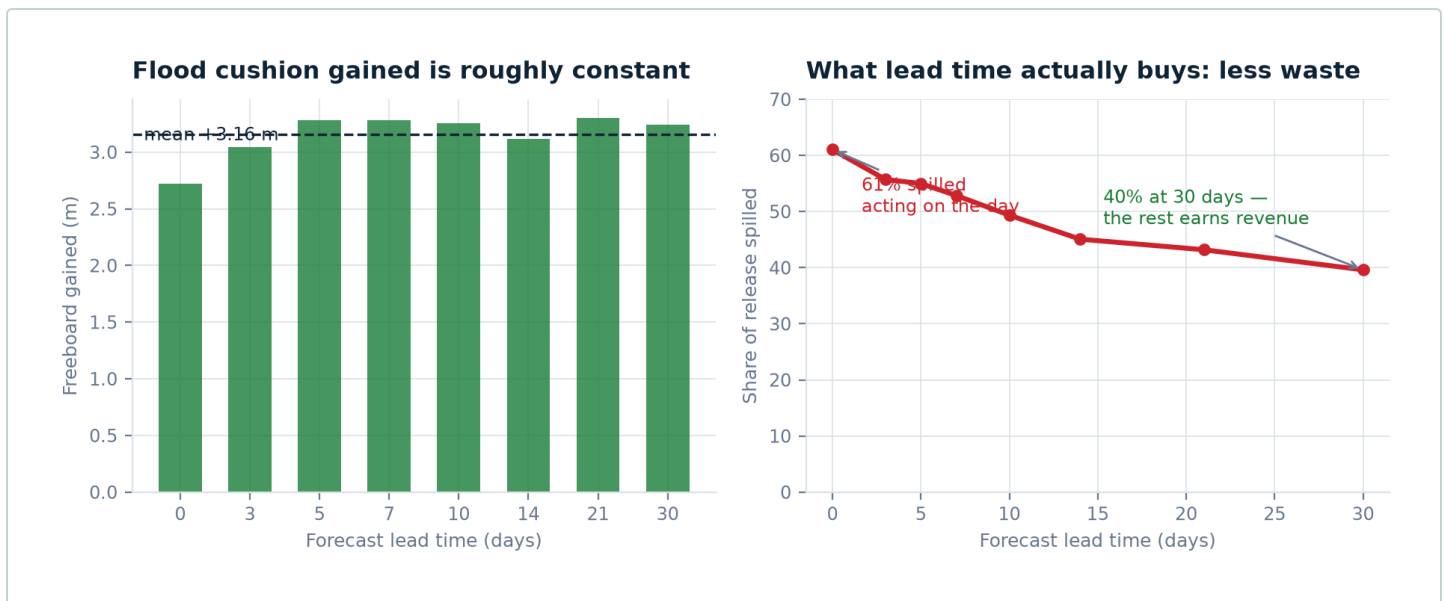


Figure 6 – flat cushion, falling waste. The left panel is the result people expect to slope upward and does not. The right panel is where lead time actually pays, and it is the reason a forecast-driven system earns money rather than costing it.

07 – THE COUNTERFACTUAL

The rule was already right

Given only the physics and the objective – no knowledge of KSEB's operating manual – the policy search converged on a target level of 728.43 m. KSEB's published rule level is 728.50 m. The optimiser independently rediscovered the existing operating rule to within **seven centimetres**.

The recommendation is not **change the rule**. The rule is right. What is missing is a system that **acts on it against a forecast**, early enough that acting is cheap.

Replayed over the observed October 2021 inflow record, that schedule holds the reservoir near 728.46 m instead of climbing to 731.54 m – about **3.08 m of additional freeboard** at the moment it matters – while generation revenue rises rather than falls, because the same volume of water passes through the turbines in higher-tariff hours instead of over the spillway at whatever hour the crisis dictates.

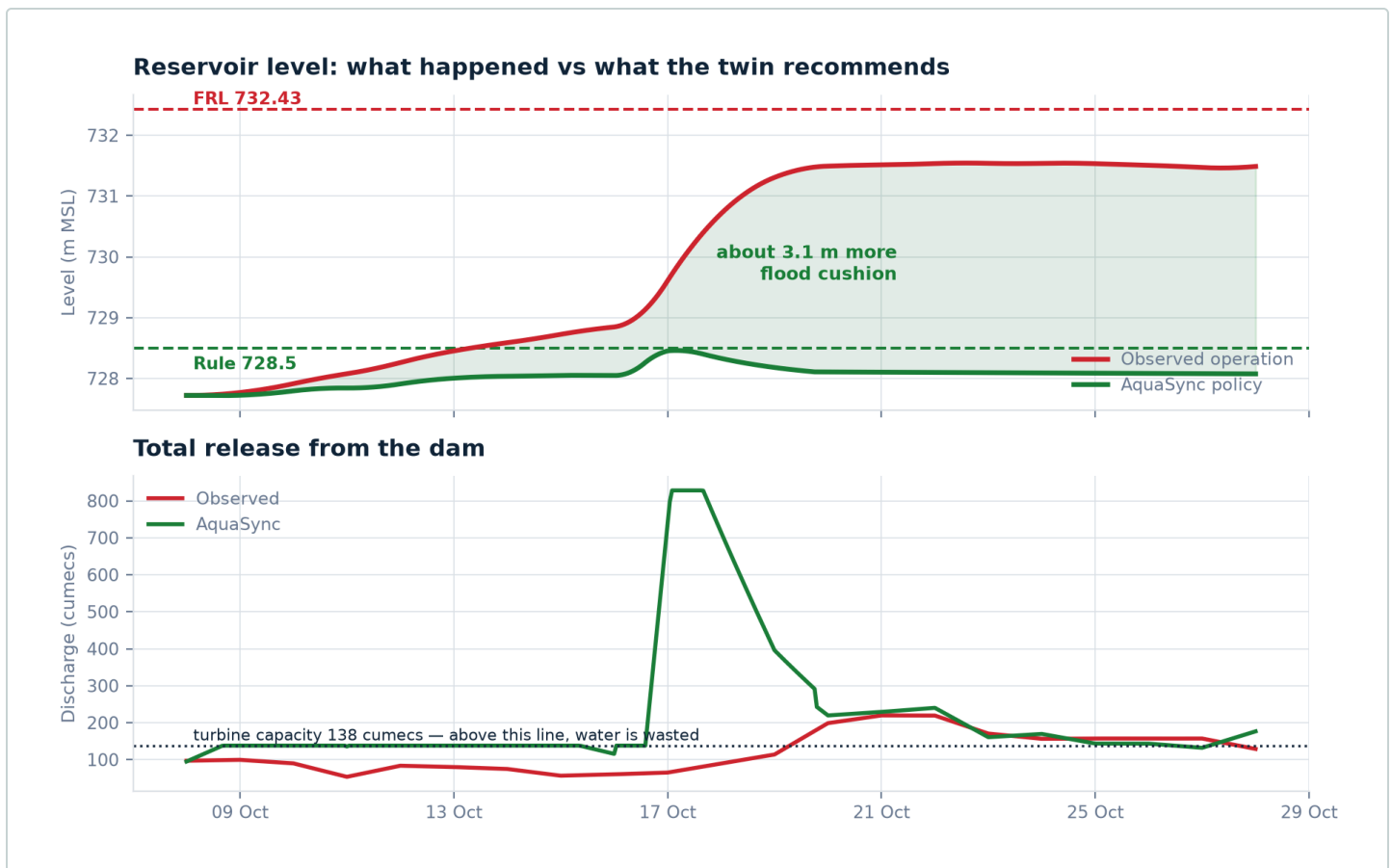


Figure 7 – the same storm, scheduled. The shaded band is the cushion. The lower panel shows what it costs: a deliberate release above turbine capacity on 17 October, taken while the reservoir is still low, instead of an involuntary one taken later at 730.95 m. Above the dotted line water is spilled and earns nothing – the optimiser crosses it once, on purpose.

Hardware in the loop

A two-tank recirculating scale model with a motorised sluice gate. The twin computes a policy and the physical gate executes it – and the demonstration is only convincing if you can break it on purpose.

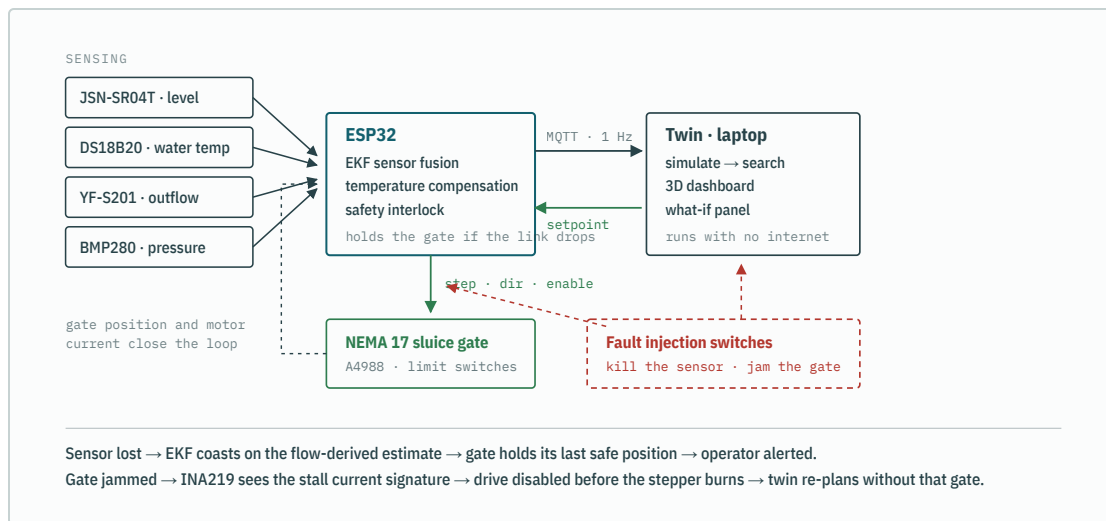


Figure 8 – the rig, and how it stays in control. The dashed red paths are the demonstration that matters: with a sensor cut or the gate jammed on demand, the system detects the fault, degrades cleanly and keeps controlling.

COMPONENT	SPECIFICATION	WHY THIS PART
ESP32-WROOM-32	38-pin, 240 MHz	38-pin, not 30 – the extra ADC pins are needed. Runs fusion and interlock locally, so the rig survives a dropped link.
JSN-SR04T v3.0	25–450 cm, ± 1 cm	Transducer on a cable, so it survives splash. A dry HC-SR04 handles the tailwater tank.
DS18B20	± 0.5 °C, 1-Wire	Not optional. Sound speed drifts 0.6 m/s per °C – uncompensated, that is millimetres of invented level.
NEMA 17 + A4988	4.2 kg·cm, 1/16 step	Drives the rack-and-pinion sluice. A stepper has no position feedback, hence the limit switches.
SPDT limit switches	×2 on GPIO 34/35	Input-only pins with no internal pull-ups – external ones required. Homing and end-stops.
YF-S201	1–30 L/min	Measures what actually left, rather than what the gate was told to do.
INA219	± 3.2 A, I^2C	A stepper missing steps against an obstruction has a current signature. This is how the twin knows a gate is jammed rather than merely slow.
SX1278 LoRa	433 MHz, ~5 km	433 MHz is the Indian ISM allocation. The 868 MHz modules recommended online are the European one.

SAFE BY DESIGN

The rig **does** close the loop onto a motor – because it is a 30 cm acrylic tank. The deployed system never would. A named officer approves every release, and that boundary does not move once the demo is over.

09 – BUILT AND WORKING

What runs today

This is not a proposal. Every component below is written, merged, and reproducible from the repository – and every number in this document comes out of running it.

COMPONENT	STATE	RESULT IT PRODUCES
Data ingestion – KSEB bulletin	Complete	18 reservoirs, six years of daily records, one schema.
Validation layer	Complete	Physical-plausibility gate on every row, with a printable quality report per dam.
Level-storage calibration	Complete	$\beta = 1.348$, $r^2 = 0.9957$, MAE 17 Mm ³ on 1,836 rows.
Reservoir mass balance	Complete	Hourly integration; replays October 2021 to 0.30 m mean error.
SCS-CN rainfall-runoff	Complete	Antecedent-moisture shift with a triangular unit hydrograph.
Muskingum + Muskingum-Cunge routing	Complete	Automatic sub-reaching for numerical stability on long reaches.
Tidal backwater	Complete	Kochi harmonics, fully offline – no API, no network.
Hydropower and tariff model	Complete	Hill-diagram efficiency; 743 MW at rated flow against 780 MW nameplate.
Objective function and policy search	Complete	Deterministic grid search under a grid-offtake constraint.
Rule-curve baseline	Complete	A faithful model of current practice – the benchmark to beat.
October 2021 flagship scenario	Complete	~3 m additional cushion with higher generation revenue.

COMPONENT	STATE	RESULT IT PRODUCES
Lead-time study	Complete	Spill share falls 61% → 40% across the forecast horizon.
3D twin dashboard	Complete	Three.js, no build step, WebSocket telemetry with smoothing.
Rig BOM and firmware	Complete	Costed to ₹6,150, with pin map, sensor fusion and safety interlock.
Documentation set	Complete	Architecture, data sources, validation report, and a 15-page dossier.

Two properties make this reproducible rather than merely finished. **Every figure regenerates** – no number in this document was typed by hand, and if the upstream data changes the figures change with it. And the **simulation core has no framework dependencies**: `twin/` imports nothing from `api/`, so the physics can be tested on its own, and it is.

The whole demonstration runs with the network cable pulled. Tide prediction, the twin, the policy search and the dashboard are all offline-capable by design, because a tool that only works when the internet is up is not a disaster-management tool.

10 – WHERE THIS GOES

Connecting the water bodies

Two dams is where the argument starts, not where it ends. The same failure repeats wherever reservoirs share a river, and the fix is the same: schedule them as one system.

Section 03 showed the missing edge – Idukki and Idamalayar releasing onto the Periyar on the same day, each decision locally correct and the sum unowned. That edge is not unique to those two dams. It exists between every pair of reservoirs whose water eventually meets, and Kerala has a great many of them.

The substrate is already here. The ingestion layer does not load two dams. It loads **eighteen**, across six districts, six years of daily bulletins each, through the same validated schema and the same quality gate. Every one of them is a node waiting for an edge.

RESERVOIRS INGESTED	DISTRICTS	COMBINED LIVE STORAGE	DAILY RECORDS EACH	COVERAGE
------------------------	-----------	--------------------------	-----------------------	----------

What co-optimisation adds is not more water released. It is the same water released at different **times**. Two release pulses that arrive together make a peak; the same two pulses staggered by a few hours make a plateau the channel can carry. Nothing is sacrificed – not storage, not generation – only the coincidence is removed.

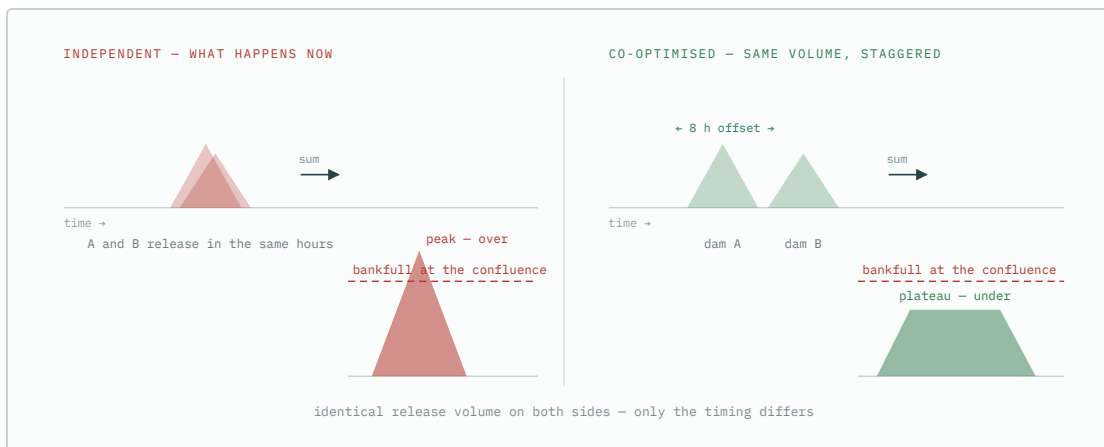


Figure 9 – what coordination actually buys. The shaded area under each pulse is the water released, and it is the same on both sides. Coincidence is what turns two safe releases into one unsafe one, and timing is the only variable that has to change to remove it.

The engineering is smaller than it sounds. The routing layer is already a directed graph – **RiverNetwork** – so a second dam is a node and a reach, not a rewrite. Extending the policy search from one node to many is the next step, and it is the highest-value item on the roadmap.

Beyond the Periyar, the same structure scales: reservoirs and confluences are nodes, reaches are edges, and the tidal boundary at each river mouth is the terminal constraint. A statewide layer is the same model with more nodes – not a different model.

The expansion path is already mapped and each step is small. Cascade co-optimisation across Idukki and Idamalayar comes first, because the routing graph and both datasets are in place. From there the same model absorbs CWC fifteen-minute telemetry to move from daily hindcast to live nowcast, field-deployed ESP32 nodes to fill the gaps between official gauges, and Malayalam last-mile alerting through LDMC and Kudumbashree networks so the warning reaches the people it is for. Every one of those is an addition to a working system, not a rebuild of it.

BUILT FOR ADOPTION

The system is designed to run **in shadow mode** from day one: alongside real operations for a full monsoon, logging what it would have advised and publishing the comparison. That gives an operator a complete performance record before a single recommendation is ever acted on – which is exactly how a decision-support tool earns its place in a control room.

11 – REFERENCES

What this was built on

Fourteen open-source projects and datasets were acquired, verified and read before a line of the twin was written. Every URL was checked with a real HTTP request and its status code recorded.

The verification was not ceremony. Every part number, dataset claim and citation behind this project was independently re-checked before it was relied on – which is why the data corrections in §05 were caught early enough to shape the design rather than late enough to undermine it.

RESOURCE	KIND	WHAT IT CONTRIBUTED
amith-vp / Kerala-Dam-Water-Levels	dataset	The KSEB daily bulletin for 18 reservoirs – the primary data source, and the feed whose 11% corrupt block drove the validation layer.
NOAA-OWP / t-route	repo	Operational reference for Muskingum–Cunge routing on a river network, including the sub-reach stability treatment.
google-research / flood-forecasting	repo	State of the art in operational flood forecasting at national scale; the benchmark this project's approach is measured against.
rtc-tools – Deltares	repo	Real-time control of water systems; the closest existing analogue to the decision layer.
pywr	repo	Water resource network modelling – the node-and-edge formulation behind the cascade design in §10.
Deltares / pyflwdir	repo	Flow-direction and catchment delineation from a DEM; the route to a full basin topology.

RESOURCE	KIND	WHAT IT CONTRIBUTED
neuralhydrology	repo	LSTM rainfall–runoff benchmarks – read, and the reason the runoff model here is physical rather than learned.
AI4Water	repo	ML tooling for hydrological time series; surveyed for the runoff-validation work.
mhpi / dMC-Juniata-hydroDL2	repo	Differentiable Muskingum–Cunge – the calibration approach if gauge data becomes available.
jsosa / LFPtools	repo	LISFLOOD-FP preparation tooling; the intended path to 2D inundation.
cordmaur / Sentinel1-Flood-Finder	repo	SAR flood-extent mapping – the route to satellite validation of flood extent.
pastas	repo	Time-series analysis for hydrological signals; used in assessing the level–storage fit.
opendatakerala / lsg-kerala-data	dataset	Local self-government boundaries – the geography the last-mile alerting layer would target.
KSEB live bulletin feed	dataset	The live reservoir endpoint, for the operational-nowcast path.

The corpus was gathered by thirteen agents in three phases: six researching one domain each, six adversarially re-checking every URL independently and rejecting anything they could not confirm, and one synthesising the integration paths into the build plan this project followed. The full inventory regenerates with `python scripts/acquire.py -index-only`.

SOURCE CODE

github.com/Am4l-babu/aqua-sync

Simulation core, firmware, dashboard, the research index, and every script that regenerates the figures on this page.

PHOTOS & VIDEO

Project media folder

Rig build photographs, demonstration video, and additional diagrams.

SIMULATION

SERVICE & INTERFACE

EMBEDDED

Python · NumPy · SciPy · Pandas.
No framework imports in the twin
core.

FastAPI · WebSocket telemetry ·
MQTT / Mosquitto · Three.js and
WebGL, no build step.

ESP32 · PlatformIO · C++ · LoRa
SX1278 at 433 MHz for off-grid
fallback.

DATA

KSEB daily bulletin, 18 dams, Aug
2020 → Aug 2026. IMD / Open-
Meteo forecast. INCOIS tide
harmonics.

Every figure on this page regenerates from **scripts/make_figures.py** against committed data. Nothing is hand-entered. If the upstream data changes, the figures change with it.